

Problem Set 4

Applied Stats/Quant Methods 1/Nour Kebbi/ Answer Sheet

Due: November 27, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in **R**, please include the code you used to get your answers. Please also include the **.R** file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 27, 2025. No late assignments will be accepted.

Question 1: Economics

In this question, use the **prestige** dataset in the **car** library. First, run the following commands:

```
install.packages(car)
library(car)
data(Prestige)
help(Prestige)
```

We would like to study whether individuals with higher levels of income have more prestigious jobs. Moreover, we would like to study whether professionals have more prestigious jobs than blue and white collar workers.

- (a) Create a new variable **professional** by recoding the variable **type** so that professionals are coded as 1, and blue and white collar workers are coded as 0 (Hint: **ifelse**).

```
1 #Part A
2 #head(Prestige)
3 Prestige$professional <- ifelse(Prestige$type == "prof", 1, 0) #creating
  the variable and setting it as follows: if type is Prof then
  Professional Variable is 1 else it is 0
4 #head(Prestige)
5 #table(Prestige$professional)
```

- (b) Run a linear model with **prestige** as an outcome and **income**, **professional**, and the interaction of the two as predictors (Note: this is a continuous $\times$ dummy interaction.)

```
1 #Part B
2 #running linear model in a continuous dummy interaction
3 interact_reg <- lm(prestige ~ income + Professional + income:Professional
  , data = Prestige)
4 summary(interact_reg) #seeing the results
```

The prestige is estimated to be 21.14 if income is equal to 0 and for non-professionals. On average, for non-professionals, the prestige will increase 0.003 for every 1 unit of income increase. On average the prestige will increase 37.78 for every 1 unit of professional (type is Prof) increase when income is 0. On average, for professionals, the effect of income on prestige will decrease by 0.002

- (c) Write the prediction equation based on the result.

```
1 #Part C
2 #prediction equation
3 prestige <- 21.14 + (0.003 * income) + (37.78 * Professional) - (0.002*(
  income*Professional))
```

- (d) Interpret the coefficient for **income**.
On average, for non-professionals, the prestige will increase 0.003 for every 1 unit of income increase.
- (e) Interpret the coefficient for **professional**.
On average the prestige will increase 37.78 for every 1 unit of professional (type is Prof) increase when income is 0.
- (f) What is the effect of a \$1,000 increase in income on prestige score for professional occupations? In other words, we are interested in the marginal effect of income when the variable **professional** takes the value of 1. Calculate the change in $\hat{y}$ associated with a \$1,000 increase in income based on your answer for (c).

```

1 #Part F
2 prestige_F = 21.14 + (0.003 * 1000) + (37.78 * 1) - (0.002*(1000))
3 prestige_F #the prestige will be 59.92
4 marginal_effect_calcul = 0.003 - 0.002
5 marginal_effect_calcul #0.001 per unit of income
6 marginal_effect <- marginal_effect_calcul * 1000
7 marginal_effect #1 increase in prestige

```

The predicted prestige for a professional with 1000 USD income is 59.92. The marginal effect of a 1000 usd increase in income for professionals is an increase of 1 unit to prestige

- (g) What is the effect of changing one's occupations from non-professional to professional when her income is \$6,000? We are interested in the marginal effect of professional jobs when the variable **income** takes the value of 6,000. Calculate the change in $\hat{y}$ based on your answer for (c).

```

1 #Part G
2 prestige_G = 21.14 + (0.003 * 6000) + (37.78 * 1) - (0.002*(6000))
3 prestige_G #the prestige will be 64.92
4 marginal_effect_calcul_g = 37.78 - 0.002 * 6000
5 marginal_effect_calcul_g #25.78 per professional

```

The predicted prestige for a professional with 6000 USD income is 64.92. The marginal effect of changing from non-professional to professional at 6000 usd income is an increase of 25.78 units in prestige.

Question 2: Political Science

Researchers are interested in learning the effect of all of those yard signs on voting preferences.¹ Working with a campaign in Fairfax County, Virginia, 131 precincts were randomly divided into a treatment and control group. In 30 precincts, signs were posted around the precinct that read, “For Sale: Terry McAuliffe. Don’t Sellout Virginia on November 5.”

Below is the result of a regression with two variables and a constant. The dependent variable is the proportion of the vote that went to McAuliffe’s opponent Ken Cuccinelli. The first variable indicates whether a precinct was randomly assigned to have the sign against McAuliffe posted. The second variable indicates a precinct that was adjacent to a precinct in the treatment group (since people in those precincts might be exposed to the signs).

Impact of lawn signs on vote share	
Precinct assigned lawn signs (n=30)	0.042 (0.016)
Precinct adjacent to lawn signs (n=76)	0.042 (0.013)
Constant	0.302 (0.011)

Notes: $R^2=0.094$, $N=131$

- (a) Use the results from a linear regression to determine whether having these yard signs in a precinct affects vote share (e.g., conduct a hypothesis test with $\alpha = .05$).

```
1 coef <- 0.042
2 se <- 0.016
3 t_value <- coef/se
4 p_value <- 2*(1-pnorm(abs(t_value)))
5 p_value #0.008
```

0.008 is less than 0.05 (pvalue<0.05) then we reject the null hypothesis that the yard signs significantly affect vote share.

- (b) Use the results to determine whether being next to precincts with these yard signs affects vote share (e.g., conduct a hypothesis test with $\alpha = .05$).

¹Donald P. Green, Jonathan S. Krasno, Alexander Coppock, Benjamin D. Farrer, Brandon Lenoir, Joshua N. Zingher. 2016. “The effects of lawn signs on vote outcomes: Results from four randomized field experiments.” Electoral Studies 41: 143-150.

```

1 #part B
2 coef_2 <- 0.042
3 se_2 <- 0.013
4 t_value_2 <- coef_2/se_2
5 p_value_2 <- 2*(1-pnorm(abs(t_value_2)))
6 p_value_2 #0.0012

```

Since $0.0012 < 0.05$, we reject the null hypothesis that being adjacent to precincts with yard signs has no effect on vote share

- (c) Interpret the coefficient for the constant term substantively.
The intercept is 0.302. This is the expected vote share if there are no signs neither in the precincts or adjacent to it.
- (d) Evaluate the model fit for this regression. What does this tell us about the importance of yard signs versus other factors that are not modeled?
R squared is 0.094. this means that the model explains only 9.4 percent of the variation in the vote share across projects. The remaining 90.6 percent of the variation is explained by other factors that are not included in the model. This means that while yard signs have an effect, they are not a major determinant of election results.